

Experimental study on the effect of grouting volume on the tensile performance of offset semi-grouted couplers

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ABSTRACT

Research results have shown that the mechanical properties of sleeve grouting connections are related to the diameter of the rebar, the sleeve, the grout, the temperature and the loading method, but research on the factor of grout volume needs to be further developed. Therefore, this paper considers the effects of three grouting volumes (50%, 75% grouting and 100% full grouting) on the tensile properties of the joint under the offset condition of the rebar. A total of 22 uniaxial tensile tests on semi-grouted steel sleeve connection specimens were carried out, mainly to study the damage morphology, load-displacement relationship, stiffness and the strain change law on the surface of the rebar and the sleeve under different grouting amount, etc. The stress mechanism of the specimens in the tensile condition was analysed, and the effective bond length calculation model was proposed with reference to the steel-concrete bond strength research theory, and the formula for calculating the maximum bond stress related to the effective anchorage length was obtained. Studies have shown that the grouting amount affects the load bearing capacity and stiffness, and that the quality of the specimen joint is reliable when the grouting volume reaches more than 75% of the design grouting volume.

1. Introduction

On 31 August 2015, “The Action Programme to Promote the Production and Application of Green Building Materials” issued by the Ministry of Industry and Information Technology and the Ministry of Housing and Urban-Rural Development pointed out that we should vigorously develop prefabricated concrete buildings, the research on prefabricated components have become a hot issue concerned by scholars [1–3]. The safety of the prefabricated structure design is focused on components and the connection, semi-grouting sleeve connection is mainly used for building structure in the wall, column and column-plate beam bridge project, the connection between the column and foundation; Thanks to the good performance, the advantages of low cost, simple construction, the semi-grouting sleeve connection has a broad prospect of application and research space [4–6].

Studies have shown that the tensile properties of sleeve grouting joints are affected by such factors as steel bar grade, steel bar diameter, steel bar anchorage length, transition splicing of coupled steel bars, sleeve structure, grouting material strength, temperature and load form [7–11]. Several conclusions from domestic and foreign research are

listed: 1) The configuration of sleeve end affects the connection performance, it is recommended that the ratio of the inner diameter of the end sleeve to the diameter of the steel bar be 1.3, the ultimate strength increases with the rebar diameter increase [12]; Zheng YF [13] developed a new grouting sleeve made of standard seamless steel tube, and the paper concluded that increasing the number and height of ribs inside the sleeve can effectively improve the bond strength; 2) The bond strength between grout material and rebar increases with the increase of grout material strength [14]. The bond strength is proportional to the square root of the grouting material's compressive strength [15,16]. Changing the anchorage length of rebar can improve the bond strength more effectively than increasing the strength of grouting material, and the minimum anchorage length of rebar can be changed by changing the internal form of sleeve [17]; 3) Temperature has influence on the ultimate load and ductility of sleeve grouting joint [18]. After experienced high temperatures of 200°C, 400°C and 600°C respectively, the decreasing degree of bond strength and bearing capacity becomes larger, the peak temperature affects the strength, strain, ductility coefficient and equivalent elastic modulus of the sleeve grouted couplers after fire [15]. After 600°C, 800°C and 1000°C, the bond strength

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decreases [10]; 4) The bonding strength and ductility between rebar and grout can be improved effectively by adding short cut fiber into grout, and the minimum anchorage length of rebar can be reduced [19], the use of ultra-high performance concrete (UHPC) as grouting material can greatly affect the bond strength, ductility and failure mode of the large-diameter steel-sleeve grouting joint [20]; 5) Under cyclic loading, the bond strength of sleeve grouting connector decreases compared with that of static single pull down [21–23].

Grouting material plays the role of transferring load between steel bar and sleeve [24,25], but in practical projects, it is inevitable that grouting quantity is insufficient. However, there are few research datas on whether the amount of grouting affects the performance of sleeve connection. Therefore, more research needs to be carried out in combination with the actual engineering to form a complete research system, so as to improve the engineering design and construction basis.

This paper simulated the actual grouting state and carried out single-pull test to study the influence of grouting amount on the mechanical properties of the sleeve grouting connection. Based on the basis of the existing research results and test datas, relevant suggestions for engineering design and construction are put forward in the paper. The research can provide theoretical support for the application of prefabricated vertical component connection in engineering.

2. Experimental investigation

The test scheme was designed to study the tensile capacity and failure mode of the sleeve grouting connection under different grouting amount levels, and to improve the relevant research on the influencing factors of the sleeve connection.

2.1. Specimen design

In practical engineering, there may be construction positioning error, which may lead to steel bar bias. Therefore, this test is aimed at the semi-grouting sleeve joint in the bias state of reinforcement, and the study variables are grouting amount and reinforcement diameter. The schematic diagram of the specimens is shown in Fig. 1.

According to the current standard Technical specification for grout sleeve splicing of rebars (JGJ 355–2015) and Technical specification for mechanical splicing of steel reinforcing bars (JGJ 107–2016), semi-grouted sleeve connection specimens with bias reinforcement were designed and single pull tests were carried out. The diameters of the connecting steel bar selected are 14 mm, 16 mm and 20 mm which are commonly used in engineering. Grouting material strength grade is C60, the size of 40 mm × 40 mm × 160 mm test block grouting material compressive strength were tested, and the measured strength is 85 MPa; The grouting amount $\frac{L_1}{L_0}$ is taken in accordance with the reference [11]. When the diameter of reinforcement is 20 mm, the grouting amount is

50 %, 75 %, and 100 % of full grouting; When the diameter of reinforcement is 14 mm and 16 mm, the grouting amount is 75 % and 100 % of full grouting grouting respectively. In order to ensure the accuracy of grouting amount, the specimens were placed vertically on the fixed frame for grouting, and a position marking line was drawn on the outside of the sleeve. When the grouting material reached the marking line, the grouting amount was measured. According to the research purpose and significance, 7 groups of working conditions were designed, with at least 3 specimens in each group. In fact, 4 specimens were made in each group. When there was no reinforcement fracture outside the joint or bond failure caused by reinforcement slipping out, the test of the fourth specimen was tested. A total of 22 specimens of semi-grouted sleeve joint were subjected to unidirectional tensile tests, and one specimen under the condition of d14-3/4 was found to have rebar fracture at thread of the rebar. Therefore, a total of 4 specimens were tested in this group, and the average values of d14-3/4-2, d14-3/4-3 and d14-3/4-4 specimens were taken to calculate the bearing capacity, stiffness and strain under this condition. The details of specimen design are shown in Table 1.

2.2. Material properties

The grouting material and steel bar of the same batch were tested, and the strength grade of grouting material was C60. According to the specification Test Method of Cement Mortar Strength (GB/T17671), the corresponding compressive test was carried out, and the measured compressive strength of grouting material test block (40 mm × 40 mm × 160 mm) under the same condition was 85 MPa; The specimen sleeve adopts section steel sleeve. The corresponding sleeve models of steel bars with different diameters are GT14, GT16 and GT20. The sleeve lengths of the three types are 156 mm, 174 mm and 211 mm respectively. The anchorage lengths of steel bars at the grouting end in full grouting state are 112 mm, 128 mm and 160 mm respectively; The brand plate of the reinforcement is HRB400, the yield strength and ultimate strength of 14 mm steel bar are 544.5 MPa and 729 MPa respectively; The yield strength and ultimate strength of 16 mm diameter reinforcement are 529.1 MPa and 708.6 MPa respectively. The yield strength and ultimate strength of 20 mm diameter reinforcement are 488.1 MPa and 679.9 MPa respectively.

2.3. Test device and loading method

As shown in Fig. 2, the 1000kN MTS tensile testing machine was used to complete the tensile test.

According to A.1.4 of JGJ 107-2016 and the reference [9]. According to article 5.0.6 of Technica specification for grout sleeve splicing of rebars (JGJ 355-2015), the loading rate was get in Appendix A.14 of JGJ 107-2016 for details. The recommended static loading rate is 0.05lc

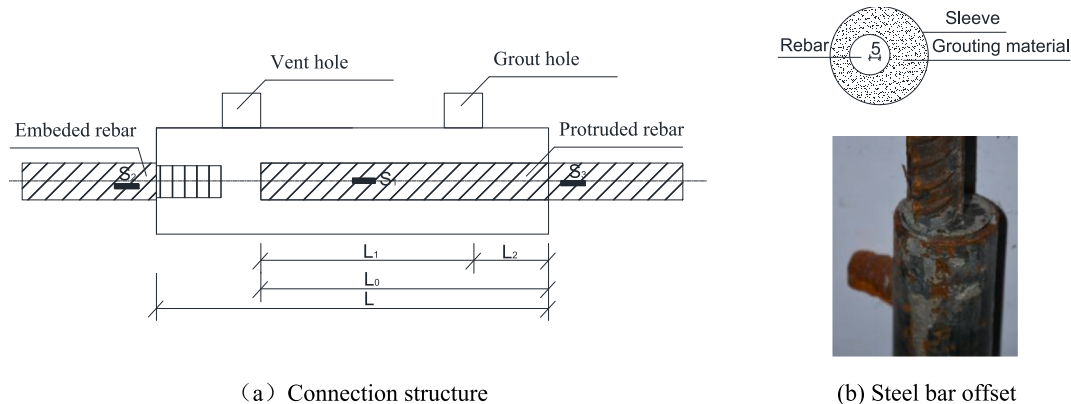


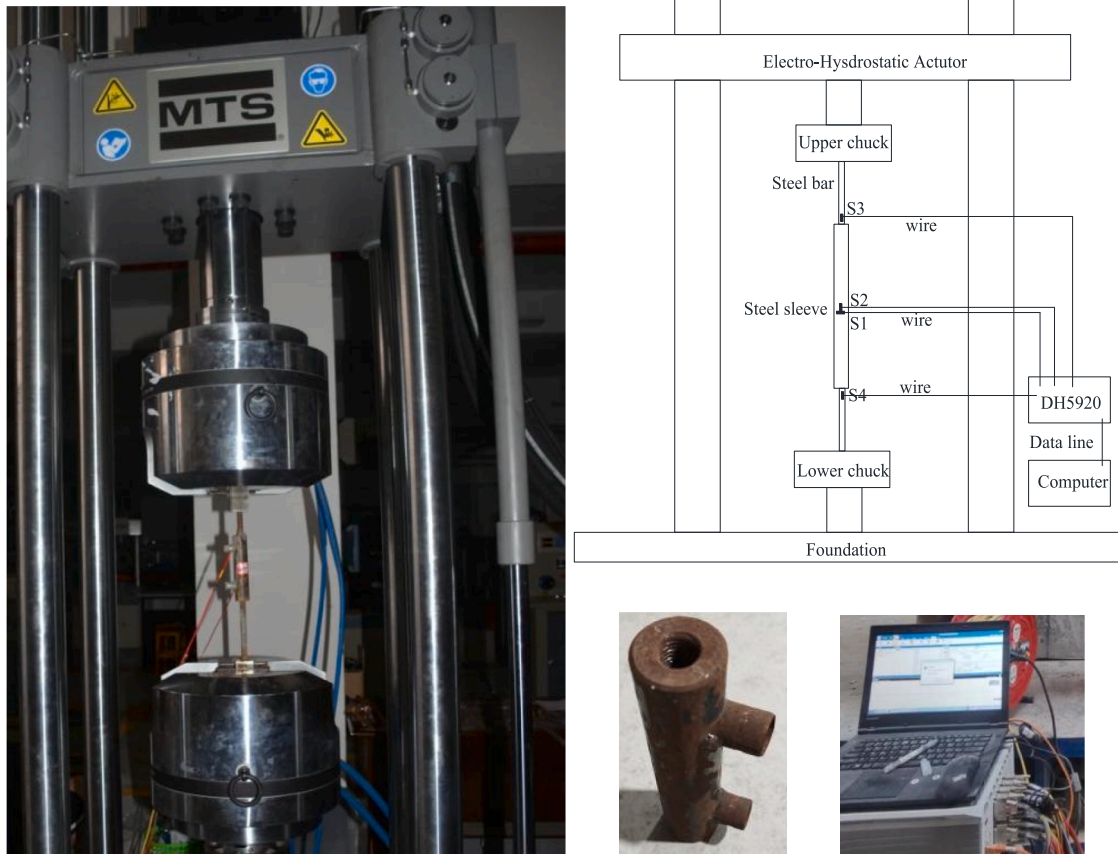
Fig. 1. Schematic diagram of the half-grouting sleeve connection.

Table 1

Test piece details and test results.

Specimen number	Anchorage length L_1 (mm)	Measured grouting volume	The yield load P_y (kN)	The average $\bar{P}_y$ (kN)	$\frac{ P_y - \bar{P}_y }{\bar{P}_y}$	Ultimate load P_u (kN)	The average $\bar{P}_u$ (kN)	$\frac{ P_u - \bar{P}_u }{\bar{P}_u}$	Total elongation	Failure mode
d14-1-1	112	100.0 %	71.6	78.5	8.79 %	90.4	98.3	8.04 %	6.1 %	A
d14-1-2	112	100.0 %	83.6		6.50 %	102.9		4.68 %	6.3 %	A
d14-1-3	112	100.0 %	80.4		2.42 %	101.5		3.26 %	7.9 %	A
d14-3/4-1	82	73.2 %	71.8	–	–	96.2	–	–	7.3 %	C
d14-3/4-2	83	74.1 %	72	70.4	2.27 %	99.6	98.1	0.31 %	8.8 %	A
d14-3/4-3	83	74.1 %	72.2		2.56 %	98.3		1.63 %	6.9 %	A
d14-3/4-4	83	74.1 %	67.1		4.69 %	96.5		1.84 %	8.4 %	A
d16-1-1	128	100.0 %	110.4	110.6	0.18 %	142.3	143.0	0.71 %	8.4 %	A
d16-1-2	128	100.0 %	110.2		0.36 %	143.5		0.35 %	6.6 %	A
d16-1-3	128	100.0 %	111.3		0.63 %	141.5		1.06 %	7.5 %	A
d16-3/4-1	92	71.9 %	109.4	109.4	0.00 %	139.3	141.6	0.87 %	6.3 %	A
d16-3/4-2	93	72.7 %	110.3		0.82 %	140.7		0.36 %	8.9 %	A
d16-3/4-3	95	74.2 %	108.6		0.73 %	144.9		1.23 %	8.4 %	A
d20-1-1	160	100.0 %	141	141.5	0.35 %	199.7	199.2	0.25 %	8.7 %	A
d20-1-2	160	100.0 %	141.2		0.21 %	194.2		2.51 %	8.3 %	A
d20-1-3	160	100.0 %	142.3		0.57 %	203.6		2.21 %	7.8 %	A
d20-3/4-1	121	73.8 %	138.5	140.8	1.63 %	190.4	196.1	2.91 %	8.3 %	A
d20-3/4-2	118	75.6 %	143.6		1.99 %	204		4.03 %	6.9 %	A
d20-3/4-3	119	74.4 %	140.4		0.28 %	193.9		1.12 %	7.3 %	A
d20-1/2-1	81	50.6 %	142.1	140.0	1.50 %	187.3	181.7	3.08 %	4.3 %	B
d20-1/2-2	73	45.6 %	137.8		1.57 %	166.4		8.42 %	3.6 %	B
d20-1/2-3	86	53.8 %	140.1		0.07 %	191.4		5.34 %	4.4 %	B

Note: d and the following numbers in the specimen number represent the diameter of steel bar, in mm. The middle numbers 1, 3/4 and 1/2 represent 100% full grouting, 75% grouting and 50% grouting respectively. The ratio of length L_1 to L_0 was taken in the experiment. The last number represents the sequence of specimens in the same group; Failure mode A is the reinforcement broken, B is the bond failure of the reinforcement sliding out, C is the fracture failure of the reinforcement at the thread, and P_{sy} is the average value of the measured yield load at the yield stage.

**Fig. 2.** Test set up.

/min, $L_c = 500 \text{ mm} \sim 1000 \text{ mm}$, and the rate range is $0.416 \sim 0.833 \text{ mm/s}$. Therefore, the intermediate value of 0.6 mm/s is taken as the loading rate.

Firstly, the loading was carried out at a rate of 1 kN/s under force control to $0.6f_{yk}A_s$ and then unloaded to 0 kN to eliminate the effect of the initial deviation caused by the installation, and then the loading was carried out at a rate of 0.6 mm/s under displacement control to failure. The $0.6f_{yk}A_s$ of the specimens with the diameter of 14 mm , 16 mm and 20 mm were 50.3 kN , 63.8 kN , and 92.0 kN .

Three strain gauges were arranged on the radial of the sleeve and the reinforcement near the sleeve at both ends and the strain was collected by DH5920. The strain data of the reinforcement at the damaged end is selected for data analysis. The load–displacement datas were obtained from MTS, that was calibrated before the test to ensure the accuracy and reliability of the data.

3. Experimental results and discussion

3.1. Failure modes

Table 1 and Fig. 3 show the failure modes of the specimens. In the tests involved in this paper, there are three types of failure: 1) Mode A: Bar fracture outside the joint. This pattern disruption occurred in most of the specimens, when the amount of grout is enough, the bond force generated at the rebar/mortar interface exceeded the ultimate tensile capacity of rebars; subsequently, the reinforcing bars were ruptured by tensile loading as shown in Fig. 3(a). Due to the weakened circumferential restraint effect of the end sleeve on the grouting material, a small amount of mortar collapsed at the end of the failure; 2) Mode B: Bond failure. Three specimens with 50% grouting amount had B failure mode, when the anchorage of reinforcement is insufficient, the bond strength between reinforcement and grouting material is lower than the ultimate strength of reinforcement itself, and then the bond failure occurs; For this pullout failure, the specimens failed with the yielding of reinforcing bars. 3) Mode C: Bar fracture at thread. Only d14-3/4-1 occurred for the disruption of this pattern, for the semi-grouted sleeve connection, if the reinforcement at the thread is weakened too much, it will easily lead to the fracture of the reinforcement at the thread of the joint.

Therefore, in the actual project construction, special attention should be paid to two points: First, when the semi-grouting sleeve connection is adopted, the steel bar should be processed strictly in accordance with the design requirements, and the thread should not be overprocessed to avoid weakening the steel bar section too much; The second is to ensure enough grouting, the construction of the grouting nozzle is not easy to leak, it is recommended to adopt the method of secondary grouting, along with the detection of the density of grouting.

3.2. Analysis of bearing capacity and displacement relationship

Fig. 4 shows the load–displacement curves of specimens. The mechanical properties results of sleeve grouting joints considering grouting amount are summarized in Table 1. It can be seen from Table 1 and Fig. 4:

When the amount of grouting is about 50% , the reinforcement pull out failure will occur (Fig. 4 (b)), when the grouting amount is 75% and 100% , the tensile failure of the steel bar will occur in the specimens with three diameters (Fig. 4 (a)).

Except for the specimens where the reinforcement fracture occurred at the thread and the specimens where the reinforcement slip failure occurred, the total elongation of the other specimens all met the quality requirements of the first-class joint in the specification. When the grouting amount reaches more than 75% , it can meet the quality requirements of the first grade joint.

All the specimens of d20-1/2 were damaged by rebar pulling out. This is because the contact surface of the reinforcement and grouting material is too small, mechanical bite force is insufficient; There is a difference between the actual grouting amount and the theoretical value when machining the specimen, so the bearing capacity of each specimen is different. Therefore, grouting quantity must exceed 75% of theoretical design value during construction.

Due to the particularity of the sleeve grouting connect structure, the load - displacement curve is not completely similar to a single reinforcement. After the beginning of yield, the grouting material cracks, and the relative displacement between the reinforcement occurs, But the steel bar is still constrained, so the yield platform is moving slightly higher.

It can be seen from Table 1 and Fig. 4:

The average yield load and ultimate load of d14-1 are 78.5 kN and 98.3 kN , respectively. The average yield load and ultimate load of d14-3/4 are 70.4 kN and 98.1 kN , respectively, which are 10.3% and 0.2% lower than that of d14-1; The average yield load and ultimate load of d16-1 are 110.6 kN and 143.0 kN , respectively. The average yield load of d16-3/4 are 109.4 kN , 1.08% lower than that of d16-1, and the average ultimate load of d16-3/4 is 141.6 kN . It was 0.98% lower than that of d16-1; The average yield load and ultimate load of d20-1 are 141.5 kN and 199.2 kN , respectively; the average yield load and ultimate load of d20-3/4 are 140.8 kN and 196.1 kN , respectively, 0.49% and 1.56% lower than that of d20-1. The average yield load and ultimate load of d20-1/2 are 140.0 kN and 181.7 kN , respectively, which are 10.06% and 8.79% lower than that of d20-1.

With the decrease of grouting amount, the yield load and ultimate load show a decreasing trend on the whole; When the amount of

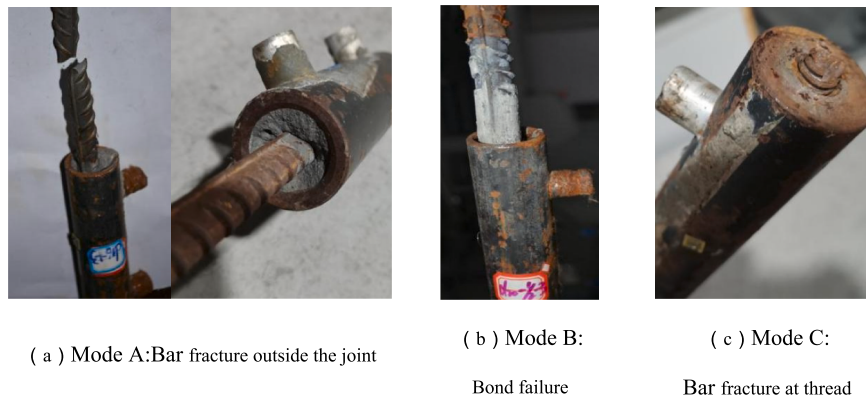


Fig. 3. Failure mode of grout sleeve splices.

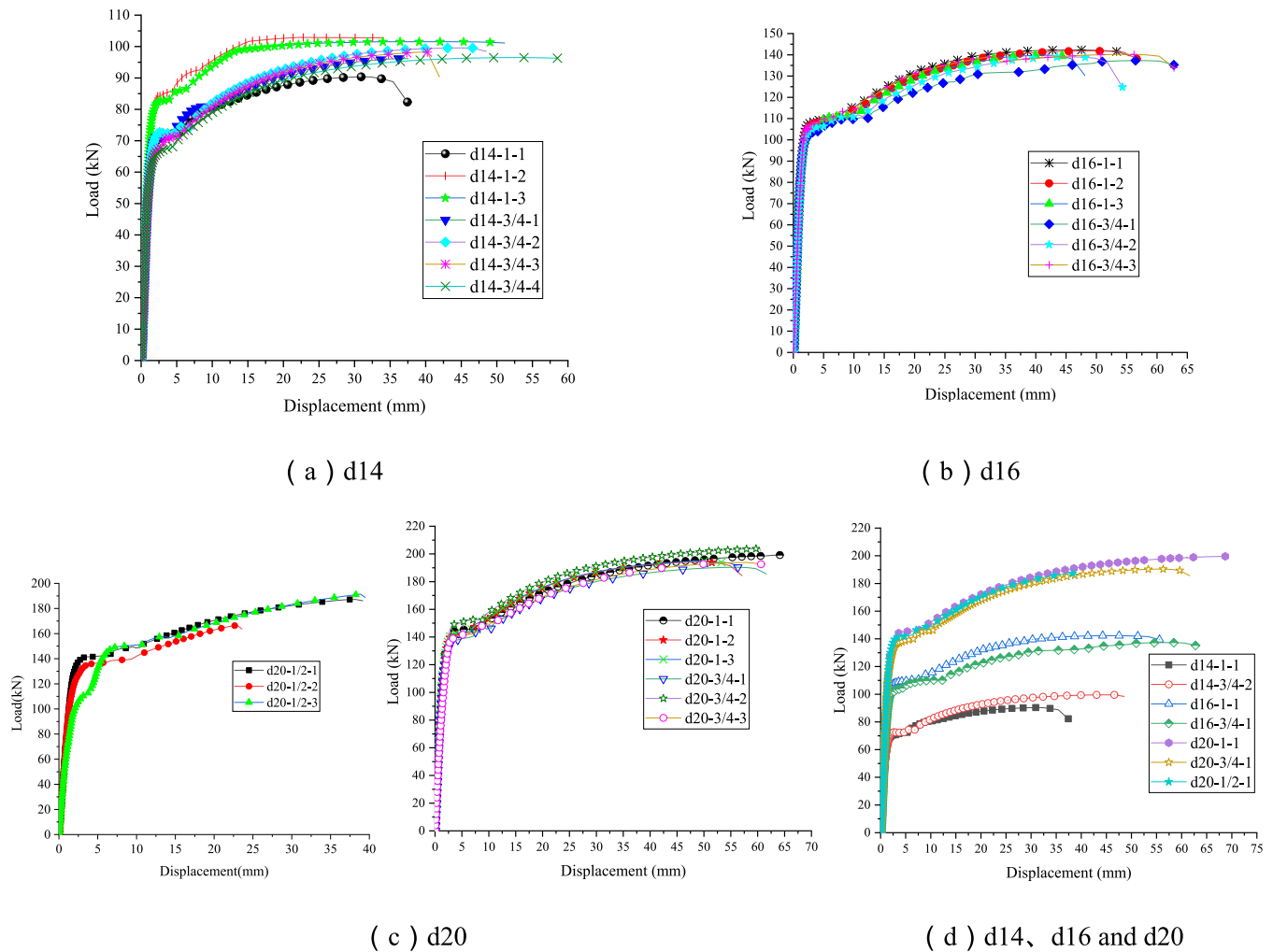


Fig. 4. The load-displacement relation curve.

grouting is less than 50 %, bond failure occurs, and the bearing capacity depends on the bond strength between reinforcement and grout material. When the grouting amount is more than 75 %, the yield load and ultimate load change slightly, and the bearing capacity of the sleeve grouting connection specimen depends on the diameter of rebar when the bar fracture occurs. This indicates that with the increase of grouting amount, the bearing capacity of sleeve grouting connection specimens does not increase indefinitely. There is an effective action range between grouting and grouting material, and the bearing capacity does not increase when the range is exceeded.

As can be seen from Fig. 4, except for the specimens d20-1/2 in which the steel bar pull out, the load-displacement curves of all the other specimens are similar, and they have experienced the elastic stage, yield stage, strengthening stage and neck stage. There are obvious yield platforms, the larger the diameter of rebar is, the greater the ultimate displacement and the energy dissipation capacity is. The load displacement curve of the specimen d14-3/4-1 with fracture failure at the thread is similar to that of the steel bar fracture. The steel bar reaches the yield state and goes through the strengthening stage and finally suffers the neck failure; The failure specimen d20-1/2 reached the yield state but not the limit state, and the energy dissipation capacity and ultimate displacement are obviously lower than those of the specimens d20-1-1 and d20-3/4-1, which were analyzed to be due to the

insufficient grouting amount and the small friction force, gripping force and mechanical bite force between grouting material and steel bar.

To sum up, the bond failure should be avoided in engineering, and the quality of grouting should be controlled during construction to ensure the anchorage length of rebar and fullness of grouting, so as to prevent the failure of reinforcement sliding out in actual engineering.

3.3. Analysis of relationship between ultimate bearing capacity and grouting amount

Fig. 5 shows the relationship between the bearing capacity of specimens with different steel bar diameters and grouting amount. It can be seen from Table 1 and Fig. 5:

- 1) The ultimate bearing capacity of d14-1 is 98.3kN, and the ultimate bearing capacity of d16-1 and d20-1 are 143.0kN and 199.2kN respectively, which increase by 45.5 % and 102.6 % compared with d14-1. With the increase of steel bar diameter, the joint bearing capacity of semi-grouted sleeve connection specimens increases;
- 2) For the specimens with 100 % full grouting and 75 % grouting, the bearing capacity of the three reinforcement diameters all show similar regularities, that is, the bearing capacity of the reinforcement with the same diameter is basically the same under the condition of full grouting and 75 % grouting. Therefore, the failure of the specimen begins when the reinforcement outside the joint is pulled, and

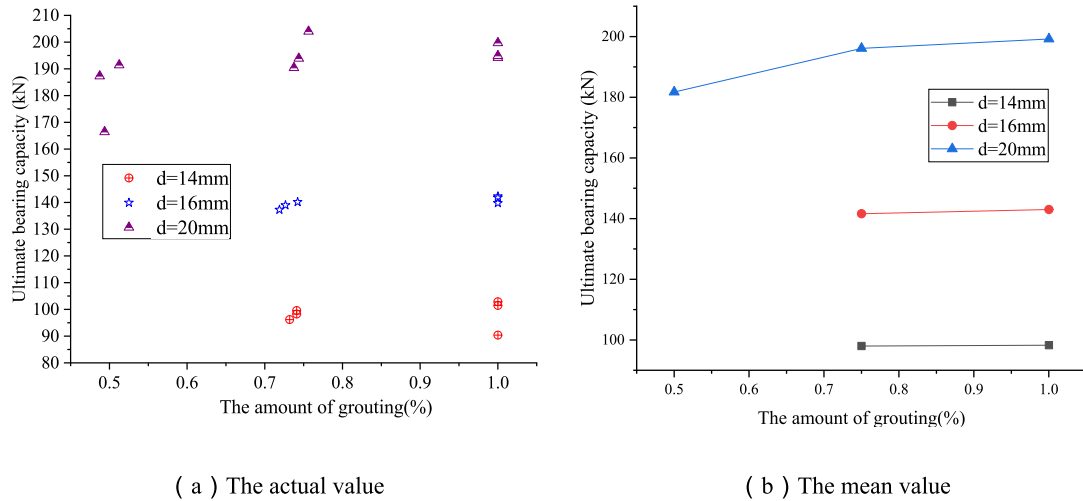


Fig. 5. Relationship between ultimate bearing capacity and grouting amount.

the bearing capacity depends on the performance of the reinforcement.

- 3) When $d = 20$, the bearing capacity of d20-1/2 is about 6 % lower than that of d20-3/4 and d20-1. Therefore, When it occurs bonding damage between reinforcement and grouting material, the bearing capacity of steel bar depends on the size of the bond between grout material and steel bar.

3.4. Analysis of bearing capacity and strain relationship

When the steel bar outside the joint is broken, the bearing capacity of the specimen is basically the same with the same steel bar diameter and different grouting amount. Therefore, three representative specimens were selected to describe the relationship between bearing capacity and strain of semi-grout sleeve connection specimens with different steel bar diameters (see Fig. 6).

Table 2 lists the load values when the reinforcement and sleeve strains reach the maximum strain. The longitudinal strain of the sleeve is increased linearly with the increase of the load before the yield stage, however, it begins to decrease slightly after the yield stage, the main reason is that the grouting material around the embedded rebars cracked after the yield stage, which led to the longitudinal shrinkage of the sleeve. In the whole tensile process, the sleeve is not yield, which meet the strength requirements.

Table 2

Relationship of maximum strain and load.

Specimen number	Strain gauge position	Maximum strain	Ratio of the corresponding load to the ultimate load
Specimen 14-1-4	Sleeve	2933.2	89.1 %
	Rebar	6518.2	90.1 %
Specimen 16-1-1	Sleeve	994	95.9 %
	Rebar	5792.6	84.3 %
Specimen d20-1-3	Sleeve	1007.1	77.4 %
	Rebar	5450.9	74.1 %

Before reaching the yield load, the strain of reinforcement presents a linear-nonlinear relationship with the load, after the yield load, there is an obvious yield stage, which is due to the grout material cracking, and growing of the relative slip between the steel bar and the grouting material, and the constraint of the steel bar is relatively smaller; For the specimen d14-1-1, When the load reaches 89.1 % of the ultimate load value, the sleeve strain reaches the maximum value of 2933.2 $\mu\epsilon$, When the load reaches 90.1 % of the ultimate load value, the maximum reinforcement strain reaches 6518.2 $\mu\epsilon$; For the specimen d16-1-1, When the load reaches 95.9 % of the ultimate load value, the maximum sleeve strain reaches 994 $\mu\epsilon$, When the load reaches 84.3 % of the ultimate load value, the maximum reinforcement strain reaches 5792.6 $\mu\epsilon$; For the specimen d20-1-3. When the load reaches 77.4 % of the ultimate load

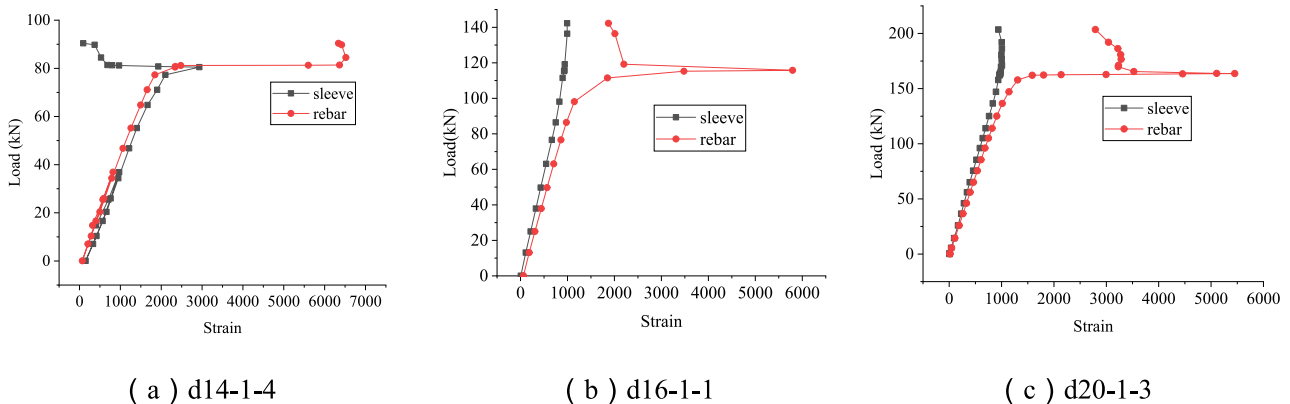


Fig. 6. Relationship between capacity and strain.

value, the sleeve strain reaches the maximum value of $1007.1\mu\epsilon$. When the load reaches 74.1 % of the ultimate load, the reinforcement strain reaches the maximum value of $5450.9\mu\epsilon$. It can be seen that the load conditions corresponding to the maximum strain of reinforcement and sleeve of different specimens are different. The analysis shows that the mortar volume in the sleeve is affected by the cracking of the mortar when the reinforcement is yielding, but the influence degree of the mortar on the strain of the sleeve is different because of the variation degree of the mortar; When the specimens with different steel bar diameters were destroyed, the steel bar strain was about $6000\mu\epsilon$ and the sleeve strain was between $900\mu\epsilon$ and $3000\mu\epsilon$.

According to the above analysis, due to the particularity of the structure, the strain turning point of the reinforcement and sleeve in the sleeve grouting joint is different from that of a single reinforcement and sleeve.

3.5. Stiffness analysis

As shown in Fig. 7, the stiffness N of the semi-grout sleeve connection specimens is obtained by linear fitting with the data of linear stage load and displacement, and the calculation results are listed on Table 3.

According to Fig. 7 and Table 3, the average stiffness of d14-1 is 73.9 kN/mm , and that of d14-3/4 is 71.9 kN/mm , 2.71 % lower than that of d14-1; The stiffness of d14-3/4-1, where the reinforcement fracture occurred at the thread, was 66.7 kN/mm , 9.74 % and 7.04 % lower than that of d14-1 and d14-3/4, respectively. Based on the bearing capacity data in Table 1, it can be seen that the bearing capacity of the specimens with reinforcing bar fracture at the thread is basically the same as that outside the joint, which depends on the diameter of reinforcing bar, but the stiffness decreases. This is because the reinforcement section at the thread is weakened, so it is not recommended to use semi-grouting sleeve connection for members with small reinforcement diameter.

The average stiffness of d16-1 is 90.9 kN/mm , and that of d16-3/4 is 82.7 kN/mm , down by 9.02 % from d16-1; The mean stiffness of D20-1 was 98.5 kN/mm , d20-3/4 is 90.0 kN/mm , decreasing 8.63 % over d20-1, and the mean stiffness of d20-1/2 is 87.8 kN/mm , decreasing 10.9 % over d20-1.

As can be seen from Table 3, when the reinforcement diameter is the same, the stiffness value decreases as the grouting amount decreases; As the diameter of the steel bar increases, the stiffness value becomes larger. The stiffness values of 16 mm and 20 mm are increased by about 23.0 % and 33 % over the diameter of 14 mm.

Table 3

Stiffness of sleeve grouting connection.

Number	Stiffness $N(\text{kN/mm})$	The average $N(\text{kN/mm})$
d14-1-1	74.2	73.9
d14-1-2	73.2	
d14-1-3	74.3	
d14-3/4-1	66.7	71.9
d14-3/4-2	73.2	
d14-3/4-3	70.4	
d14-3/4-4	72.0	90.9
d16-1-1	92.8	
d16-1-2	89.7	
d16-1-3	90.3	82.7
d16-3/4-1	86.7	
d16-3/4-2	83.9	
d16-3/4-3	77.6	98.5
d20-1-1	95.7	
d20-1-2	95.0	
d20-1-3	104.9	90.0
d20-3/4-1	93.6	
d20-3/4-2	87.8	
d20-3/4-3	88.9	87.8
d20-1/2-1	89.6	
d20-1/2-2	86.4	
d20-1/2-3	87.3	

4. Prediction model

In this section, the prediction formulas of effective anchorage length and maximum average bond stress are presented based on the test results in this study and some available data in other literatures.

4.1. Effective anchorage length model

The anchorage length of reinforcement has a great influence on the bond stress of sleeve grouting connection [26], the test results show that for the specimens with exactly the same grouting material, steel bar and sleeve, when the grouting amount reaches 75 %, the bearing capacity does not continue to increase, so it can be considered that there is an effective bond length between steel bar and grouting material, and the anchorage length between steel bar and grouting material does not contribute significantly to the bearing capacity when the anchorage length exceeds the effective anchorage length. Studies have also shown that there is a reasonable anchorage length between reinforcement and grout material. Zhao J [27] obtained that the minimum reasonable range of anchorage length of grouting sleeve is (5d-5.5d) through tests and theoretical analysis. Wang ZQ [26] conducted tensile tests on specimens with steel bar diameters of 12 mm and 20 mm, and obtained

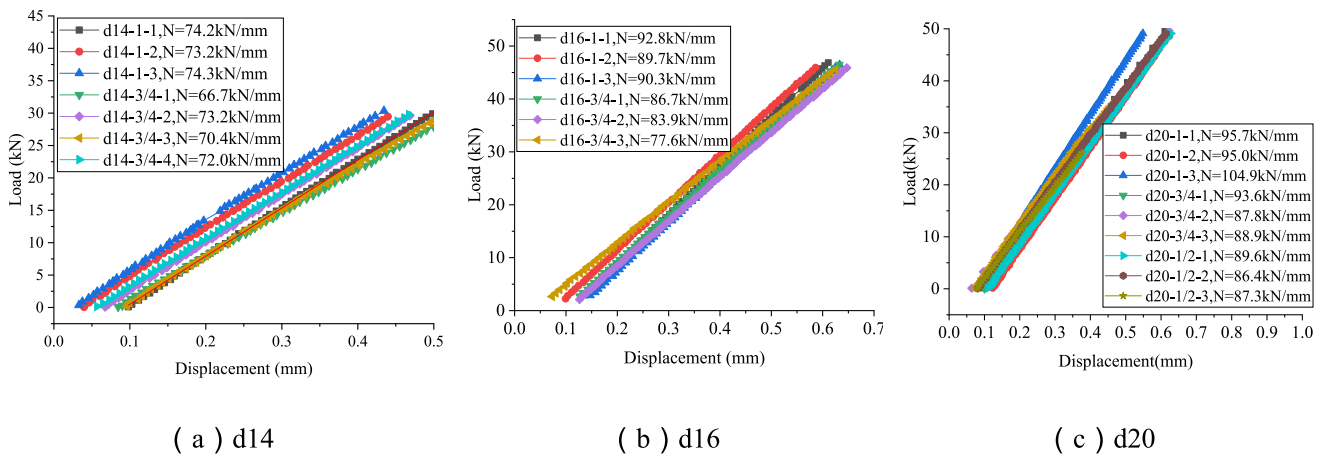


Fig. 7. Stiffness.

the critical anchorage lengths are 5.42d and 5.48d respectively. JGJ 355-2015 suggested a range of steel bar anchorage lengths of (6d-8d). Therefore, based on the existing research theory and experimental data, this paper puts forward the effective bond length model of sleeve grouting connection.

Wang ZW [26] proposed the formula of anchorage length L_e :

$$L_e = \frac{P_{sy}}{0.63df_m} \quad (1)$$

Where, P_{sy} is the measured yield force of reinforcement, kN; d is the diameter of reinforcement, f_m is the compressive strength of grouting material.

The failure mechanism of grout material is similar to concrete [27], therefore, this paper refers to the basic formula of anchoring length between steel bar and concrete in Code for Design of Concrete Structures GB50010-2010:

$$l_a = \alpha \frac{f_y}{f_t} d \quad (2)$$

Where, l_a is reinforcement anchorage strength, kN; α is the coefficient, d is the diameter of reinforcement, and f_t is the tensile strength of concrete.

It can be seen from the synthesis of Eqs. (1) and (2) that the anchorage length of steel bar is related to the diameter of steel bar and the strength of bonding material (grouting material or concrete), so the effective anchorage length formula of steel bar (3) is proposed:

$$L_e = A \frac{f_y}{f_m} d \quad (3)$$

According to the test results, the average ultimate load of d14-1 and d14-3/4 is 98.3kN and 98.1kN, and that of d16-1 and d16-3/4 are 143.0kN and 141.6kN, respectively. The average limit load of d20-1 and d20-3/4 are 199.2kN and 196.1kN respectively. It can be considered that when grouting amount reaches 75 % or more, the joint bearing capacity does not increase any more, and the anchorage length of steel bar calculated at 75 % is about 5.9d, which is close to the anchorage length recommended by JGJ355-2015 and larger than the effective anchorage length recommended by literature. Therefore, the anchorage length is considered as effective anchorage length when grouting amount is 75 %. When reinforcement diameters are 14 mm, 16 mm, and 20 mm, and the average measured anchorage length are 83 mm, 93.3 mm and 119.3 mm respectively, the theoretical lengths calculated by the

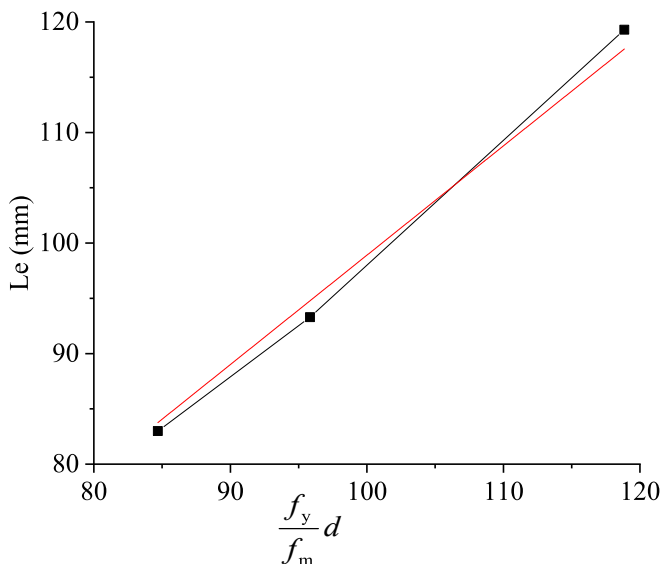


Fig. 8. Fitting plot of the effective bond length.

formula $\frac{f_y}{f_m} d$ are 89.7 mm, 99.6 mm, and 114.8 mm respectively. The effective anchorage length is calculated by linear fitting as follows Fig. 8:

$$L_e = 0.989 \frac{f_y}{f_m} d \quad (4)$$

The effective anchorage length of each specimen calculated according to Formula (1) are shown in Table 4:

It can be seen that the anchorage length calculated by Formula (1) deviates greatly from the real situation, most of the deviations are over 10 %, only individual data deviations are within 10 %. The difference between the theoretical value calculated by formula (4) and the measured anchorage length are within 10 %, therefore, the results are reliable and equation (4) can be used to estimate the effective anchorage length of the semi-grouting sleeve connection.

4.2. Maximum average bond stress model

As can be seen from the test results in Table 1, for the specimen with a diameter of 20 mm, the yield load values of the specimen with steel pulling out failure and the specimen with steel fracture are 140.8kN and 140.0kN respectively, with a difference of only 0.56 %, while the ultimate load values are 196.1kN and 181.7kN, respectively, with a difference of 7.9 %. Therefore, the ultimate load value is considered to be used to derive the maximum average bond stress.

The following formula is used to calculate the maximum bond stress when the steel bar is pulled out τ_u [9].

$$\tau_u = \frac{P_u}{\pi d L_e} \quad (5)$$

Where, P_u is the ultimate bearing capacity, kN and L_e is the measured bonding length, mm.

The calculated maximum bond stress is shown in Table 5:

Zhao YX put forward the formula of the bond force between steel bar and concrete:

$$\tau_u = \left(a + b \frac{d}{L_e} \right) \sqrt{f_{cu}} \quad (6)$$

Where, a and b are coefficients, and f_{cu} is the compressive strength of concrete cube;

Considering the bond mechanism between rebar-grout and rebar-concrete is similar, a similar bond stress calculation formula is proposed:

$$\tau_u = \left(a + b \frac{d}{L_e} \right) \sqrt{f_m} \quad (7)$$

Where, a and b are the coefficients, f_m is the compressive strength of grouting material, which is 85 MPa measured;

As it is shown in Fig. 9, a = 2.009 and b = 7.613 are obtained by linear fitting of the data. The maximum bond stress calculated from Equation (7) is listed in Table 5, and the relative deviation from Equation (5) is within 5 %. Therefore, the formula can be considered to be reliable, and the maximum bond stress between rebar and grout material can be calculated by equation (7) when bond failure occurs.

When reinforcing bar fracture failure occurs, the bearing capacity depends on the strength of reinforcing bar, when the anchorage length of reinforcing bar is equal to the effective anchorage length, the boundary state of pulling out failure and reinforcing bar fracture failure is taken to calculate the bond stress. At this point, the ultimate bearing capacity calculated according to the pull-out state is equal to the bearing capacity calculated according to the failure state of reinforcement. Referring to reference [5], it can be obtained from equations (8) and (9).

$$P_{u,b} = f_u \frac{\pi d^2}{4} \quad (8)$$

$$P_{u,b} = \tau_u \pi d L_e \quad (9)$$

Table 4

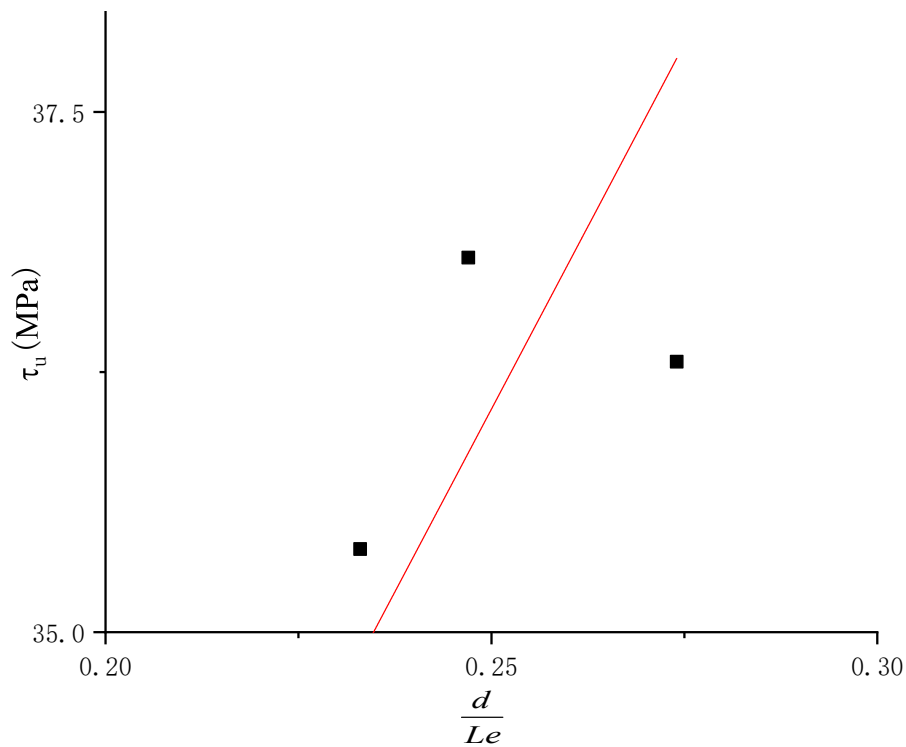
Calculation table of the effective anchorage length.

Specimen number	Anchorage length L_1 (mm)	Yield load (kN)	Equation (1) result	Relative deviation with L_1 (mm)	Equation (4) result (mm)	Relative deviation with L_1
d14-3/4-2	83	72.2	96.3	16.03 %	88.7	6.87 %
d14-3/4-3	83	72	96.0	15.71 %		6.87 %
d14-3/4-4	82	67.1	89.5	9.15 %		8.17 %
d16-3/4-1	93	109.4	127.7	37.30 %	98.5	5.91 %
d16-3/4-2	95	110.3	128.7	35.51 %		3.68 %
d16-3/4-3	92	108.6	126.8	37.77 %		7.07 %
d20-3/4-1	121	138.5	129.3	6.87 %	113.6	6.12 %
d20-3/4-2	118	143.6	134.1	13.63 %		3.73 %
d20-3/4-3	119	140.4	131.1	10.16 %		4.54 %

Table 5

Maximum bond stress of bond failure specimen.

Number	Anchorage length L_1 (mm)	Ultimate load (kN)	The average (kN)	Maximum bond stress (MPa)	Formula(5)	Relative deviation
d20-1/2-1	81	187.3	181.7	36.8	35.9	2.57 %
d20-1/2-2	73	166.4		36.3	37.8	4.00 %
d20-1/2-3	86	191.4		35.4	34.8	1.57 %

**Fig. 9.** Maximum bond stress diagram.

By integrating equations (8) and (9), the formula of maximum bond stress at fracture failure of reinforcement can be obtained:

$$\tau_u = \frac{f_u}{dL_e} \quad (10)$$

In Equations (8) ~ (10), $P_{u,b}$ is the ultimate bearing capacity of the sleeve grouting joint in the boundary state, kN; f_u is the ultimate strength of reinforcement, MPa.

To sum up, the formula of maximum bond stress is put forward:

When rebar is pulled out: $\tau_u = \left(2.009 + 7.613 \frac{d}{L_e}\right) \sqrt{f_m}$, L_e is the measured value (7).

When rebar fracture is damaged: $\tau_u = \frac{f_u}{dL_e}$ (10), $L_e = 0.989 \frac{f_y}{f_m} d$ (4).

5. Conclusion

In this paper, the factors affecting the performance of sleeve grouted coupler were studied by citing relevant domestic studies and conducting a large number of tests, and the following conclusions are listed:

In addition to the influencing factors of reinforcement, grouting material sleeve, temperature and load form determined by existing studies, the test in this paper proves that grouting amount is also an important factor affecting the mechanical properties of grouting sleeve connection, which should be drawn attention; When the grouting amount reaches more than 75 % of the design value, the joint quality is still guaranteed; When the amount of

grouting is small, the failure of reinforcement sliding out will occur and it does not meet the requirements of the code, so the amount of grouting needs to be controlled; When the construction conditions are difficult and the compaction of grouting cannot be guaranteed, sufficient amount of grouting should be guaranteed by certain methods. It is suggested that appropriate type of grouting machine and appropriate grouting gun should be selected in the actual project to reduce the leakage of grouting; The model of effective bond length and the formula of maximum bond stress are put forward:

When rebar is pulled: $\tau_u = \left(2.009 + 7.613 \frac{d}{L_e}\right) \sqrt{f_m}$, L_e is the measured value;

When rebar fracture is damaged: $\tau_u = \frac{f_u}{dL_e}$, $L_e = 0.99 \frac{f_y}{f_m} d$.

Compliance with Ethical Standards.

CRedit authorship contribution statement

Fenfang Yin: Data curation, Writing – original draft. **Shiping Yin:** Conceptualization, Supervision, Funding acquisition. **Tianhao Qin:** Resources, Visualization. **Boxue Wang:** Resources, Visualization. **Linglei Zhang:** Visualization, Funding acquisition. **Ning Yang:** Resources, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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